



Full Length Article

Towards a continuous pilot scale pyrolysis based biorefinery for production of biooil and biochar from sawdust



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ABSTRACT

Sawdust was first pyrolysed in a fixed bed batch scale reactor at temperatures between 400 °C and 600 °C. Optimum temperature for pyrolysis of sawdust was found to be 500 °C; at which, bio-oil (34.9 wt%), bio-char (38.6 wt%) and pyro-gas (26.5 wt%) were obtained. Subsequently, the reaction was scaled-up using a vertical moving bed type continuous reactor (capacity 15 kg/hr) and at 500 °C, the obtained yields of products were 41.05 wt% of bio-oil, 27.59 wt% of bio-char and 31.36 wt% of pyro-gas (mixture of non-condensable gases). Bio-oil was characterized using ^1H and ^{13}C NMR, FT-IR and GC-MS. GC-MS analysis indicates that 71 chemical compounds are present in bio-oil. The bio-char was characterized using TGA, FT-IR and SEM; these analyses showed that, it is porous in nature and contain negligible amount of volatiles. In addition, the GC analysis of pyro-gas confirms that, different gases viz. H_2 (1.59 vol%), N_2 (13.61 vol%), O_2 (2.48 vol%), CO (54.23 vol%), CH_4 (8.02 vol%) and CO_2 (24.72 vol%) are present in it.

Under optimized reaction condition, during batch pyrolysis, bio-oil yield was 34.9 wt% and bio-char yield was 38.6 wt%; whereas continuous pyrolysis gave 41.05 wt% and 27.59 wt% yields for bio-oil and bio-char respectively.

1. Introduction

Indian timber market is growing rapidly. The timber industry is a major source of sawdust as it is generated in large quantities during wood processing. As per data reported by the American Hardwood Export Council, in 2016, there were ~23,000 sawmills in India and on an average each saw mill generates ~822.4 tonnes of sawdust annually. Analyses of these data show that annually ~18.91 million tonnes of sawdust is produced in India [1]. In India, sawdust is utilized for producing light household items, toys, and furniture among others. Apart from these, substantial amount of sawdust is also used in thermal industries as loose biomass as fuel during combustion processes to produce thermal energy. Nevertheless, this causes significant risks to the environment and eventually to human health (Fig. 1).

Approximately ~5.5 million tonnes of surplus sawdust is produced annually in India [1,2]. From the waste management and circular economy point of view, this surplus sawdust can be valorized to produce fuels and chemicals. The generated bio-oil can be utilized for heating applications in furnaces, diesel engines, boilers, and gas-turbines [3]. Its blend with heavy furnace oil can also improve the combustion quality [4]. Bio-oil can be mixed with different chemical compounds to produce adhesives and resins. Furthermore, bio-oil is

pool of chemical compounds, and these compounds can be isolated and characterized for use in fine chemical and pharmaceutical industries. Porous bio-char is used as a soil amendment to improve the fertility of soil and for carbon sequestration. Bio-char is widely regarded as a catalyst for different chemical reactions and a good precursor for production of activated carbon. Furthermore, pyro-gas can be used as a gaseous fuel for thermal applications. Along this line, sawdust can be subjected to pyrolysis for the production of bio-oil and bio-char [5–19]. So far, various authors have investigated batch and continuous pyrolysis of sawdust (Table 1, Table 2) [5–19].

Many researchers have reported batch pyrolysis of sawdust; where the scale was 1.5 to 100 g (Table 1) [5–11]. Some of the research groups have reported continuous pyrolysis of sawdust (Table 2) [12–19]. Among them, only few authors have reported continuous pyrolysis of sawdust in a pilot plant scale; for instance, Olazar et al. reported non-catalytic flash pyrolysis of sawdust using a conical spouted bed reactor in a pilot plant scale (25 kg/hr) [17]. Park et al. also reported pilot scale pyrolysis of sawdust (42 kg/hr) using a continuous circulating fluidized bed system [18]. Even though many research papers are available on this subject, a majority of reports on continuous pyrolysis methods have not reported the feeding rate and residence time. This makes comparison of present study with other reported methods rather difficult

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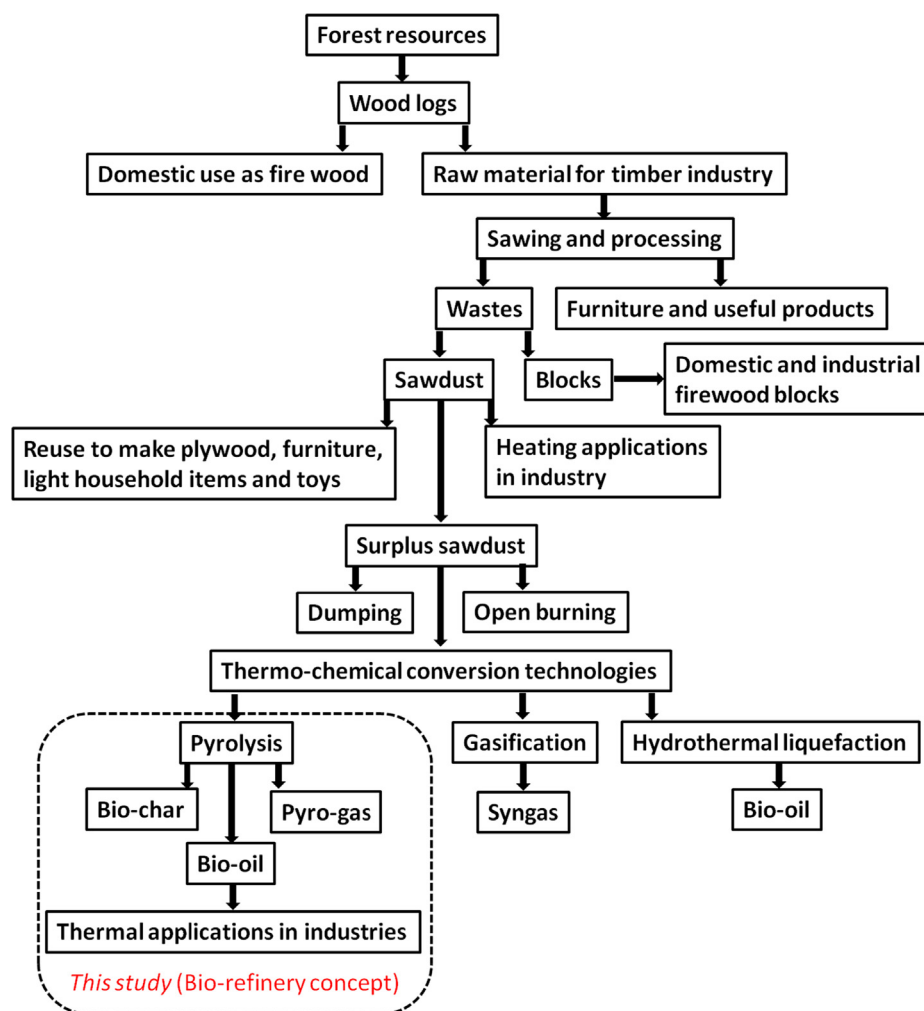


Fig. 1. Possible pathways for utilization of sawdust.

Table 1
Comparative study for batch pyrolysis of sawdust.

Sr No.	Volatile Matter (wt %)	Calorific value (MJ/kg)	Moisture (wt%)	Type of Reactor	Batch size (g)	Optimum heating rate (°C/min)	Optimum Temperature (°C)	Reaction time (min)	Yield (wt%)			Ref.
									Bio-oil	Bio-char	Pyro-gas	
1	NR*	17.9	NR*	Semi-batch	100.0	10	500	60	39.3	31.3	29.4	5
2	NR*	20.0	4.1	Free fall	1.5	NR*	550	NR*	46.8	25.5	27.7	6
3	NR*	19.05	10.0	Fixed bed	10.0	1000	500	20	44.4	23 [#]	32.6 [#]	7
4	NR*	NR*	NR*	Fixed bed	100.0	10	475	15	26	29	45	8
5	79.30	19.12	8.90	Fixed bed	100.0	10	500	NR*	42.5	27.5	24.5	9
6	80.87	18.67	3.07	Fixed bed	15.0	50	500	NR*	44.16	32 [#]	24.84 [#]	10
7	71.80	20.7	6.50	Fixed bed	20.0	50	500	NR*	54.0	25.6	20.4	11
8	76.77	17.48	8.08	Fixed bed	1000.0	13	500	50	34.9	38.6	26.5	Present study

(Table 1, Table 2) [6,9–11,13,14].

To overcome the ethical issues related to utilization of food materials as biofuel feedstocks; sawdust was valorized in this study to construct a pyrolysis based biorefinery. During the course of this research, optimization of process conditions for the bio-oil, bio-char, and pyro-gas production from sawdust was carried out initially in a batch reactor (capacity: 1 kg of sawdust). It was observed that at 500 °C maximum bio-oil yield was 34.9 wt%; whereas, 38.6 wt% bio-char and 26.5 wt% pyro-gas were generated. After optimization of sawdust pyrolysis in the batch scale reactor; subsequently, it was scaled-up in a pilot plant scale continuous system (Fig. 2). For this purpose, a vertical moving bed type

continuous reactor system was designed and developed at the Sardar Patel Renewable Energy Research Institute (SPRERI); Vallabh Vidyanagar, Anand, Gujarat, India. During continuous pyrolysis of sawdust (15 kg/hr) at 500 °C, bio-oil yield was 41.05 wt% and bio-char and pyro-gas yields were 27.59 wt% and 31.36 wt% respectively. This system is a way forward for industrial valorization of sawdust via thermo-chemical conversion based biorefinery. To the best of our knowledge, this is the first report for pilot scale intermediate pyrolysis of sawdust using a vertical moving bed type reactor.

Table 2
Comparative study for continuous pyrolysis of sawdust.

Sr No.	Volatile Matter (wt%)	Calorific value (MJ/kg)	Moisture (wt %)	Type of Reactor	Type of pyrolysis	Reactor volume (m ³)	Feeding rate (kg/hr)	Optimum Temperature (°C)	Reaction time (min)	Yield (wt%)			Ref.
										Bio-oil	Bio-char	Pyro-gas	
1	81.6	23.2	6.1	Fluidized bed	Fast	7.7×10^{-5}	0.1	500	60	67	18	15	12
2	NR*	18.07	11.5	Fluidized bed	Fast	3.4×10^{-4}	NR*	500	NR*	65	16	19	13
3	83.74	NR*	NR*	Conical spouted bed	Flash	1.12×10^{-2}	NR*	450	NR*	72	12	16	14
4	81.15	18.98	5.96	Fluidized bed	Fast	3.6×10^{-4}	0.1	500	NR*	62	12	26	15
5	91.9	NR*	9.1	Fluidized bed	Fast	1.5×10^{-3}	0.15	450	NR*	58.1	28	13.9	16
6	83.2	17.8	NR*	Conical spouted bed	Flash	NR*	25.0	480	NR*	65.80	15.40	18.80	17
7	70	17.6	9.6	Circulating Fluidized bed	Fast	NR*	42.0	500	NR*	60	15	25	18
8	90.9	17.7	7.3	Fluidized bed	Fast	1.5×10^{-3}	0.324	400	NR*	72	15	13	19
9	76.77	17.48	8.08	Vertical moving bed	Intermediate	15.91×10^{-2}	15	450	30	41.05	27.59	31.36	Present study

* NR- Not Reported
obtained from the graph of the reported publication.

2. Materials, equipment and methods

2.1. Analyses of sawdust

Sawdust is a readily available biomass in India. Sawdust used for the present study was procured from M/s Raghuvir timbers, Anand, Gujarat, India. The procured sawdust was sieved using standard sieve number 10 to get rid of wooden blocks, wood chips and remaining debris. The obtained particle size of the sawdust was ≤ 2 mm. Proximate analysis (wt%) of the sawdust was determined by following ASTM standards; namely, D-871–82 for moisture content, ASTM E-1755–01 for ash content, ASTM E-872 for volatile matter. The fixed carbon content was calculated by difference. The bulk density was determined by following ASTM E-873-06 standard. The calorific value was determined at the Thermo-Chemical Conversion (TCC) Division, SPRERI using an automated bomb calorimeter (IKA® C 5000 Cole-Parmer®, USA). All measured values are reported in the Table 3.

Volatile matter of the sawdust was ~76 wt%; whereas, the calorific value was ~17 MJ/kg (Table 3). The quantification of C, H and N were performed using an elemental analyzer (2400 Series II, Perkin Elmer, USA) at the Sophisticated Instrumentation Centre for Applied Research & Testing (SICART), Vallabh Vidyanagar, Anand, Gujarat, India. The oxygen amount present in the sawdust was calculated by difference. The carbon content in the sawdust sample was 41.38 wt% (Table 4).

2.1.1. TGA analysis

The thermal induced weight loss behavior and theoretical valorization of sawdust was analyzed using a Perkin Elmer analyzer (Pyris-1 TGA, Perkin Elmer, USA; N₂ atmosphere, Sensitivity: 0.0001 mg) at the SICART, Vallabh Vidyanagar, Anand, Gujarat, India. During the analysis, 10 mg of sawdust was heated under N₂ atmosphere at 10 °C/min from 50 °C to 1000 °C.

2.2. Experimental procedure for batch pyrolysis

Pyrolysis of sawdust was performed initially in a batch scale reactor (capacity: 1 kg of sawdust). The fixed bed batch reactor set-up was designed and developed at SPRERI, Vallabh Vidyanagar, Anand, Gujarat, India. The reactor was made up of SS304, which is enclosed by an electric resistance heating configuration (Fig. 3).

The proportional type temperature controller was used to achieve and sustain the required temperature. The axial mixing type spiral stirrer is incorporated inside the reactor for uniform mixing of sawdust during the pyrolysis. A double pipe heat exchanger was used as a primary condenser; which was maintained at ~15 °C using circulating chilled water.

Experiments in the fixed batch reactor were performed to evaluate the effect of temperature on the product yields during pyrolysis. For each batch, 1 kg of sawdust was fed into the reactor. Different operating conditions, such as, heating rate (~34 °C/min) and N₂ flow rate (100 ml/min) were kept constant throughout all sets of batch experiments. Reactions were performed at five different temperatures viz. 400–600 °C with 50 °C intervals to study the effect of temperature. Reactions were performed in triplicate to measure the accuracy of the results.

2.3. Experimental procedure for pilot plant pyrolysis

The complete pilot scale continuous pyrolysis system is comprised of different components, viz. i) feeding assembly, ii) pyrolysis reactor, iii) char extraction screw with an ashbin iv) a gas fired burner and LPG connections for reactor heating and preheating of sawdust, v) cyclone separator with a collector, vi) two sets of condenser and collector, vii) a demister pad and viii) a burner for non-condensable pyro-gas (Fig. 4).

Feeding assembly has a hopper from where biomass was fed into the jacketed allo-thermal SS316 reactor via a jacketed screw conveyor.

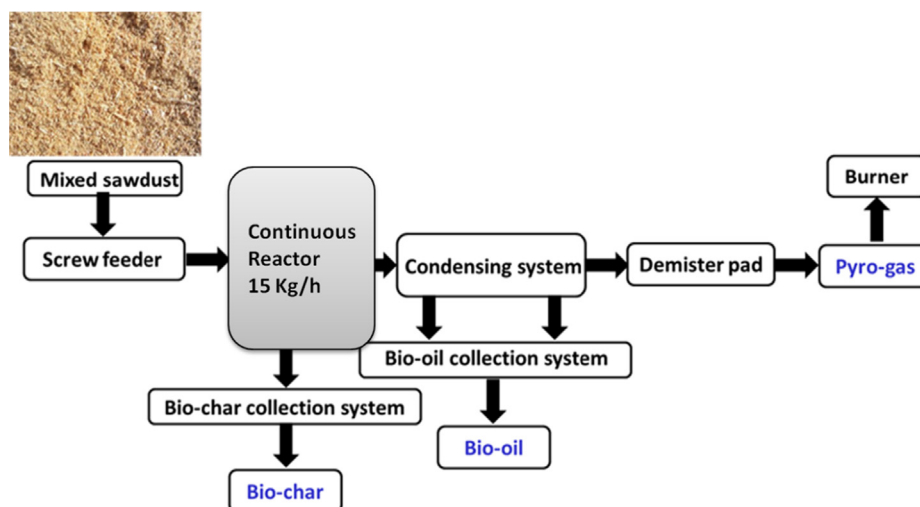


Fig. 2. Pyrolysis of sawdust for bio-oil, bio-char and pyro-gas production.

Table 3

Proximate analysis, calorific value and bulk density of sawdust.

Proximate analysis (wt%)					Calorific Value (MJ/kg)	Bulk Density (kg/m ³)
	Moisture	Ash	Volatile Matter	Fixed Carbon ^a		
w.b.	8.08	9.26	76.77	5.89	17.48	285.12
d.b.	4.98	13.15	76.02	10.83	17.33	303.40

w.b. – wet basis; d.b. – dry basis

^a By difference

Table 4

Ultimate analysis of sawdust.

C (wt%)	H (wt%)	O ^a (wt%)	N (wt%)	H/C molar ratio	Empirical equation
41.38	4.94	53.02	0.66	1.4226	CH _{1.4226} O _{0.012} N _{0.8262}

^a By difference

Objective of keeping the jackets over reactor and feeding screw conveyor is to provide space for utilization of convective heat generated through combustion of LPG into a gas fired burner (Oxilon, Surat, India; minimum gas pressure: 30/50 bar). The flue gas initially passes through reactor jacket and then diverted into the screw conveyor jacket as per the requirement to maintain necessary feedstock preheating temperature. For all sets of continuous experiments, the feedstock preheating temperature was ~110 °C. The reactor was heated and maintained at ~ 500 °C. The feed rate of preheated sawdust into the reactor was maintained at 15 kg/hr throughout all sets of experiments.

After feeding, at a specific temperature sawdust degrades it into solid and gaseous components. In order to facilitate a continuous operation, another jacketed screw conveyor was used. The char generated inside the reactor is then entrained out from the reactor using this screw conveyor and collected from a char collection bin. Purpose of jacket on this conveyor is to reduce the temperature of bio-char by circulation of chilled water.

A mixture of condensable and non-condensable gases (pyro-gas) produced during pyrolysis is passed through a cyclone separator which is provided between the reactor and primary condenser to remove the particulates and subsequently, goes through the condensers and interacts with cold water to obtain condensable gases in the form of bio-oil. Condensed liquid phase was drained out from the collection vessels that are attached with the condensers. The resulting bio-oil contains

aqueous phase, which was removed by gravity separation. The remaining non-condensable gases go through a demister pad located at the end of process line for cleaning purposes and subsequently, it was utilized for combustion via a burner which was fitted at the end of demister pad.

2.4. Characterization of pyrolysis products

2.4.1. Bio-oil characterization

Physical properties of bio-oil like solid contents, pH, viscosity, and density were analyzed at the TCC Division, SPRERI. Solid content of the bio-oil was measured by following ASTM D7579-09. pH value of the bio-oil was measured using a pH meter (pH Tutor, Eutech Instruments, Singapore). Viscosity of the bio-oil sample was measured by following ASTM D 445 and density of the sample was measured using density analyzer (DMATM 5000 M, Anton Paar, Australia).

Further characterizations of the bio-oil samples like Fourier Transform-Infrared (FT-IR), ¹H and ¹³C Nuclear Magnetic Resonance (NMR), Gas chromatography–mass spectrometry (GC–MS) were carried out at the SICART facility, Vallabh Vidyanagar, Anand, Gujarat, India. Moisture was removed from all samples via sodium sulphate (Na₂SO₄) drying. The obtained water free bio-oil samples were analyzed using NMR (¹H and ¹³C, 400 MHz FT-NMR AVANCE-III, Bruker, USA), FT-IR and GC–MS analyses. For NMR characterizations CDCl₃ was used as a solvent. The bio-oil was analyzed using GC–MS (Auto-system XL with Turbo Mass, Perkin Elmer, USA) and a capillary column PE-5MS (30 m × 0.25 mm, 0.25 μm) with helium as a carrier gas. The Fourier Transform-Infrared (FT-IR) spectrum of bio-oil sample was obtained using a Perkin Elmer analyzer (Spectrum GX FT-IR, Perkin Elmer, USA).

2.4.2. Proximate, ultimate, TGA, FT-IR and SEM analysis bio-char

TGA, calorific value, bulk density, proximate and ultimate analyses of bio-char were carried out as described for the characterization of sawdust. FT-IR characterization of the bio-char was carried out as mentioned previously in section 2.4.1. The comparison between surface morphology of raw sawdust and bio-char was performed using a Scanning Electron Microscope (ESEM XL-30, Philips, The Netherlands). The images were taken at 30 kV accelerating voltage.

2.4.3. Pyro-gas characterization

The pyro gas composition was analyzed using a Gas Chromatograph (GC) (Sigma Instruments, Vadodara, Gujarat, India) equipped with two SS packed columns namely, Molecular sieve column (3 m, 40/50 mesh) and silica gel column (2 m, 80/100 mesh). The molecular sieve column was used for identifications of H₂, O₂, N₂, CH₄ and CO, while silica gel

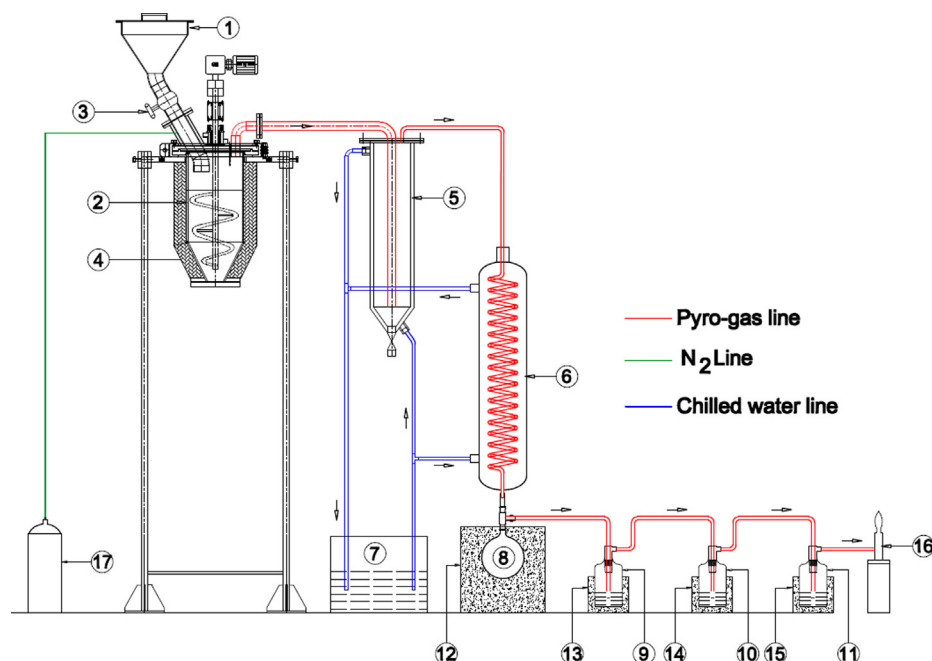


Fig. 3. Schematic representation of batch reactor. (1: Feeding hopper, 2: Reactor with stirrer, 3: Ball valve, 4: Resistance heater, 5: Primary condenser, 6: Secondary condenser, 7: Chiller setup, 8, 9, 10, 11: Collectors, 12, 13, 14, 15: Ice chambers, 16: Burner).

column was used for identification for CO_2 . The GC had a thermal conductivity detector (TCD). Column, injector and TCD temperatures were maintained at 140 °C, 200 °C and 240 °C respectively. Argon was used as an inert carrier gas throughout the experimental analyses.

3. Results and discussion

3.1. Properties of raw sawdust

For sawdust, proximate analysis, calorific value and bulk density measurements were carried out at the TCC Division, SPRERI (Table 3); In addition, ultimate analysis was carried out at SICART, Vallabh Vidyanagar, Anand, Gujarat, India (Table 4). The feedstock was found to contain ~76 wt% volatile matter and respective calorific value was ~17 MJ/kg (Table 3). Ash content of the sawdust was 9.26 wt% (Table 3). It is reported that based on the ash content biomass can be classified as low ash (< 5%), medium ash (5–10%) and high ash

material (> 10%) [20]. Similar amount of ash content results were reported for sawdust found in Gujarat, Haryana and Tamil Nadu states of India [20]. During elemental analysis of the sawdust; C, H, O and N contents (wt%) were measured (Table 4). Carbon content of the raw sawdust was found to be 41.38 wt% and empirical equation is $\text{CH}_{1.4226}\text{O}_{0.012}\text{N}_{0.8262}$ (Table 4).

The proximate and ultimate results indicate that sawdust is an ideal candidate for pyrolysis. Especially high volatile matter (~76 wt%) in the material suggests that more bio-oil yield could be obtained in the final products. Low amount of fixed carbon is more beneficial for pyrolysis as it indicates lesser yield of bio-char in the final products.

3.2. TGA

TGA is an important tool to understand the behavior of the material in an inert environment. It describes the weight loss of the material with respect to temperature in an inert atmosphere. Thermal analysis of

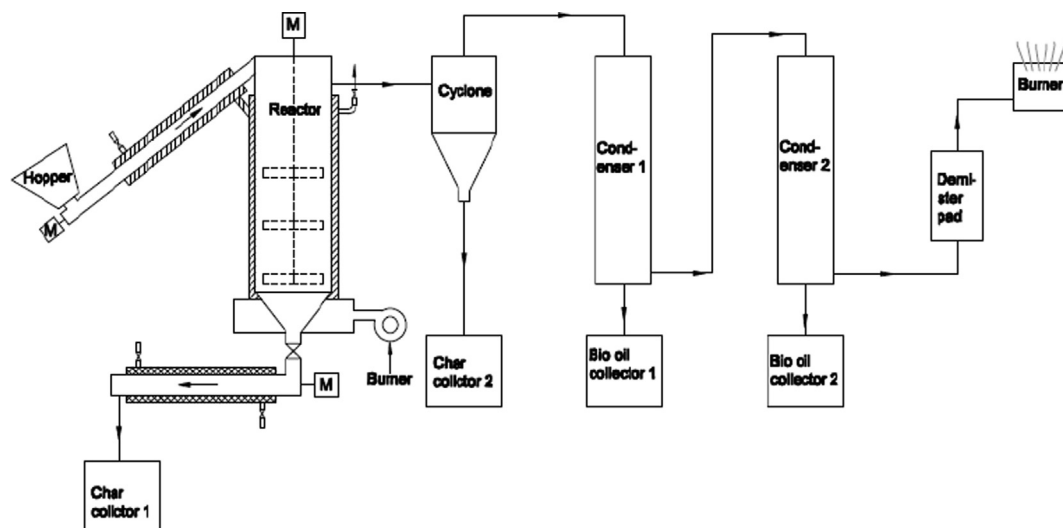


Fig. 4. Schematic representation of pilot scale continuous pyrolysis system developed at SPRERI.

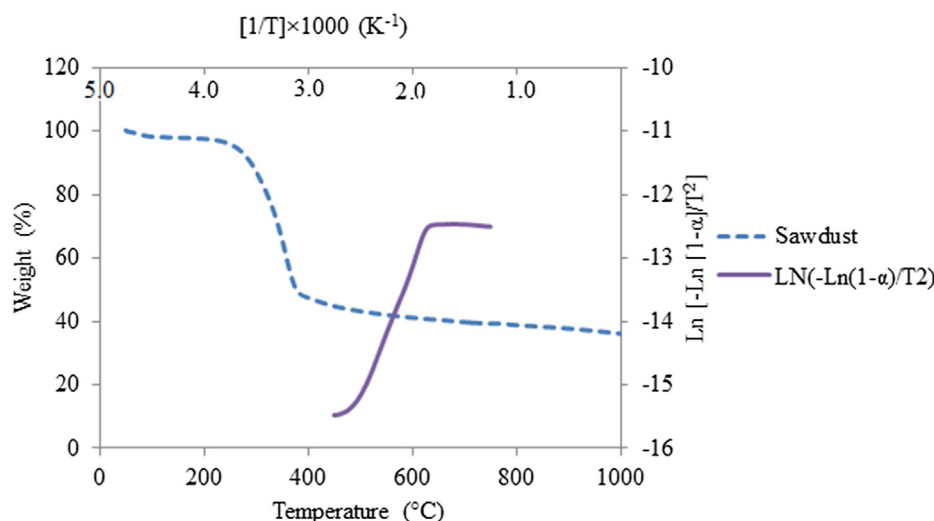


Fig. 5. TGA thermogram and Arrhenius plot of sawdust.

the sawdust is presented in the Fig. 5. Three different slopes are being observed in the thermogram. Initially 5 to 6% weight loss occurs between 0 °C and 270 °C is due to evaporation of moisture and water bounded to the sample. In the range of 300 °C to 400 °C, 39.89% weight loss occurs with highest rate. It was observed due to extraction of most of the volatiles from the sample. Subsequently, another weight loss 4.279% was observed in the range of 400 °C to 500 °C. After 500 °C, virtually negligible weight loss occurs. Generally, 320 °C to 500 °C temperature range is responsible for evolution of potential tar forming volatiles. Hence, this range can be tested in the actual pyrolysis reactors for optimization of bio-oil yield. After 500 °C degradation rate of material decreases continuously with increase of temperature and remains constant thereafter. In this range of temperature (after 500 °C), lignin from the biomass sample decomposes. Lignin is the hardest component to degrade in comparison to holocellulose components. Similar trends were observed during TGA experiments of different biomass samples such as, sawdust, cotton stalk and groundnut shell [8,20–21]. Along with weight loss analysis, activation energy for the sawdust was also determined using Arrhenius plot [22]. It was found to be 108.44 kJ/mol.

In general, thermogram helps to establish temperature range for optimum pyrolysis to achieve maximum bio-oil yield. To obtain bio-oil as a major product pyrolysis experiments were carried out between 320 °C and 500 °C as observed during TGA experiment; since this temperature range is responsible for evolution of potential tar forming volatiles. After detailed literature review it was found that the optimum temperature range for pyrolysis reaction for sawdust is 450 °C–550 °C [5–19]. Therefore, in this study the pyrolysis temperature screening was started from 400 °C.

3.3. Temperature optimization during batch pyrolysis

Pyrolysis of sawdust was performed at five different temperatures (400 °C–600 °C). At 500 °C maximum yield for bio-oil (34.9 wt%) was obtained (Table 5). At the optimum temperature (500 °C) bio-char and pyro-gas yields were 38.6 wt% and 26.5 wt% respectively (Table 5). During fixed batch pyrolysis of sawdust Kumar et al. have obtained 26% of bio-oil, 45% of pyro-gas and 29% of bio-char yield at 475 °C [8].

3.4. Continuous pyrolysis in a pilot scale continuous reactor

After performing feasibility testing and temperature optimization for higher bio-oil yield in batch; continuous pyrolysis experiments were carried out at 500 °C. At this temperature, three concurrent trials were

Table 5

Product yields obtained during batch pyrolysis at different pyrolysis temperatures.

Temperature (°C)	Product yield (wt%)		
	Bio-char	Bio-oil	Pyro-gas
400	45.0 ± 1.0	30.6 ± 0.9	24.4 ± 0.8
450	41.7 ± 0.6	32.5 ± 1.0	25.8 ± 1.6
500	38.6 ± 0.7	34.9 ± 1.0	26.5 ± 1.5
550	37.0 ± 0.6	28.5 ± 1.3	34.5 ± 0.6
600	35.6 ± 1.1	19.7 ± 0.6	44.7 ± 1.7

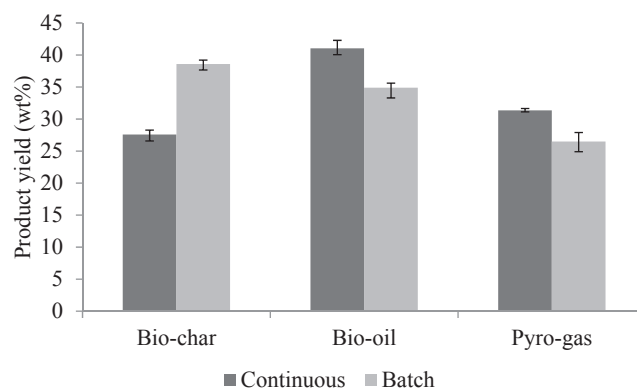


Fig. 6. Comparison of product yields between continuous and batch pyrolysis at 500 °C.

carried out and average product yield (wt%) is reported. During continuous pyrolysis at 500 °C obtained average bio-oil yield was 41.05 wt%; bio-char and pyro-gas yields were 27.59 wt% and 31.36 wt% respectively (Fig. 6). In comparison with batch pyrolysis 6.15 wt% more bio-oil yield was obtained in the continuous pyrolysis (Fig. 6). To compensate it, bio-char and pyro-gas yields decreased at that particular temperature (Fig. 6). This could be because, at the end of devolatilization process of sawdust, the generated porous bio-char can act as a catalyst and trap the generated volatiles in it [23–25]. This enhances the reactions related to secondary tar cracking. However, during continuous pyrolysis this process is less prevalent as compared to batch, due to continuous removal of bio-char from the reactor.

Table 6
Physical and fuel properties of bio-oil and furnace oil.

	Density at 26°C (kg/m ³)	Viscosity at 50 °C (cSt)	Water content (wt %)	Solid Content (wt %)	pH	Calorific Value (MJ/kg)
Bio-oil	1253	26.08	17.3	3.6	4.3	28.57
Furnace oil*	896	59.92	Nil	Nil	7.0	51.53

* Furnace oil was tested at 'Bombay Test House Pvt Ltd.', Mumbai, India.

Table 7
Proximate analysis, calorific value and bulk density of bio-char.

Proximate analysis (wt%)				Calorific Value (MJ/kg)	Bulk Density (kg/m ³)
Moisture	Ash	Volatile Matter	Fixed Carbon ^a		
5.02	25.36	22.00	47.62	23.90	293.36

^a By difference

3.5. Bio-oil characterization

3.5.1. Physical properties of bio-oil

Physical properties of bio-oil like density, water content, solid content, pH, viscosity, and calorific value were determined (Table 7). Density of the bio-oil was found to be 1253 kg/m³; it shows that, it is the dense fluid. Additionally, bio-oil is acidic in nature (pH 4.3). Physical and fuel properties of bio-oil and furnace oil were compared (Table 6). The calorific value of bio-oil (28.57 MJ/kg) is higher than sawdust (17.48 MJ/kg). In addition, calorific value of bio-oil is comparable to other liquid fuels derived from biomass, such as methanol (23.0 MJ/kg), dimethyl ether (31.7 MJ/kg) and ethanol (29.7 MJ/kg) [26]. The furnace oil has higher calorific value (51.53 MJ/kg) compare to bio-oil (28.57 MJ/kg); however, bio-oil is less viscous compare to furnace oil at 50 °C (Table 6).

3.5.2. FT-IR analysis

FT-IR is a spectroscopic technique to identify the functional groups of chemical compounds present in solid, liquid or gaseous phase. The spectrum is generated on the basis of chemical bonds available in the sample. The spectrum represents each wavelength of the light absorbed by specific chemical bonds. In this study, FT-IR was used to determine the functional groups present in bio-oil and bio-char samples.

The obtained spectrum shows the presence of major functional groups, namely, C-Br, C = C, C-H, C-Cl, C-N, O-H, N-O, and C = O in bio-oil (Fig. 7). Absorbance peaks at 3385 cm⁻¹ and 1364 cm⁻¹ suggest the O-H vibrations, which shows the presence of hydroxyl containing compounds. Absorbance peaks related to C-H bonds were appeared in the ranges of 2840–3000 cm⁻¹, 1461 cm⁻¹ and 756 cm⁻¹; these indicate availability of alkanes and alkenes. The observed peak at 1715 cm⁻¹ pointed towards the presence of carbonyl functional groups, i.e. C = O. Similarly, peaks at 1674 cm⁻¹, 790 cm⁻¹, 715 cm⁻¹ and other two peaks in the range of 885–980 cm⁻¹ indicate the existence of C = C chemical bonds in bio-oil. Two peaks in the range of 1500 to 1600 cm⁻¹ gave the indication for N-O bond. Five different absorbance peaks in the range of 1020–1350 cm⁻¹ give the hint for existence of nitrogen containing substances in bio-oil. Peaks in the range of 515–690 cm⁻¹ along with a peak at 857 cm⁻¹ could be because of the presence of halides in the sample. This spectrum shows that different classes of organic compounds such as, phenols, aliphatic alcohols, alkanes, alkenes, aromatics, ketones, aldehydes, nitrogen containing compounds, carboxylic acids and halides are present in bio-oil. The obtained FTIR results are comparable with previous investigations reported in the literature by Kumar et al. and Morali and Şensöz for sawdust bio-oil and Qureshi et al. for IR analysis of palm shell bio-oil [8,21,27].

3.5.3. ¹H and ¹³C NMR analysis

¹H and ¹³C NMR analysis is necessary for the structural

determination of chemical compounds. The ¹H NMR gives an indication about the environment of protons; whereas the ¹³C NMR indicates the environment of carbon present in the chemical compounds. The bio-oil was characterized by ¹H NMR and ¹³C NMR (Fig. 8 and Fig. 9). The ¹H NMR analysis of bio-oil reveals the presence of alkyl, aryl, allylic, ketone, and alcohol in it (Fig. 8). Along this line, Kumar et al., Yorgun et al. and Sinağ et al. have observed similar results during ¹H NMR analysis of bio-oil from sawdust [8,11,28]. ¹³C NMR analysis of bio-oil shows the presence of alkyl, aromatic, alkenes, carboxylic acids, esters and ethers in it. In addition, a small peak near 210 ppm is observed in the ¹³C NMR spectrum; it could be because of the presence of carbonyl functional group containing chemicals (Fig. 9).

3.5.4. GC-MS analysis

GC-MS is an analytical technique by which liquid chemical compounds can be identified and quantified. It is an appropriate characterization method for determination of relatively low molecular weight compounds available in the sample. Bio-oil generated from pyrolysis is complex mixture of different chemical compounds. To determine its composition, GC-MS is an important tool. Different classes of chemical compounds viz. phenols, ketones, alcohols, nitrogen containing compounds, cyclic ethers, aromatic compounds, acids, halides, aldehydes, hydrocarbons, furan and ester are found in bio-oil. This result is in line with the results reported by other research groups [14,29–32]. The GC-MS analysis revealed that in total there are seventy one different chemical compounds present in bio-oil from sawdust (see Supporting Information).

3.6. Bio-char characterization

Moisture content, ash content, volatile matter, fixed carbon, calorific value and bulk density of bio-char was characterized. During pyrolysis, volatile matter of the sawdust decreased significantly (~54 wt%); whereas after pyrolysis, higher amount of fixed carbon was found in the bio-char (47.62 wt%). In addition, the calorific value of the bio-char was 23.90 MJ/kg and bulk density was 293.36 kg/m³ (Table 7). Verma et al. and Duanguppama et al. have reported the bio-char calorific values in the similar range of 19–23 MJ/kg [10,12].

The ultimate analysis of the bio-char was determined (Table 8). The carbon content in bio-char was 74.42 wt% which is higher compared to raw sawdust (Table 8). Similar carbon content in bio-char (~72 wt%) is reported by Tinwala et al., DeSisto et al., Salehi et al. and Zhang et al. [9,13,15,33]. Lower amounts of H and O in bio-char indicate that, during pyrolysis most of the moisture and volatiles including hydroxyl group are removed from the residual product.

3.6.1. TGA analysis

Comparison of TGA thermograms of sawdust and bio-char generated from sawdust is presented in Fig. 10. In moisture removal zone (50 °C–250 °C) behavior of both the material is similar. Initial weight loss for bio-char is lesser than that of the sawdust (Fig. 12); as seen in the proximate analyses of sawdust (Table 3) and bio-char (Table 7); moisture content in the bio-char is lower than that of the sawdust. Next zone of TGA thermogram is volatile extraction; in this zone, slope for the sawdust is quite high compared to bio-char. Due to pyrolysis of sawdust, remaining volatile matter in the bio-char is very low (16.80 wt %). Therefore, the slope of its extraction is comparatively less (Fig. 10).

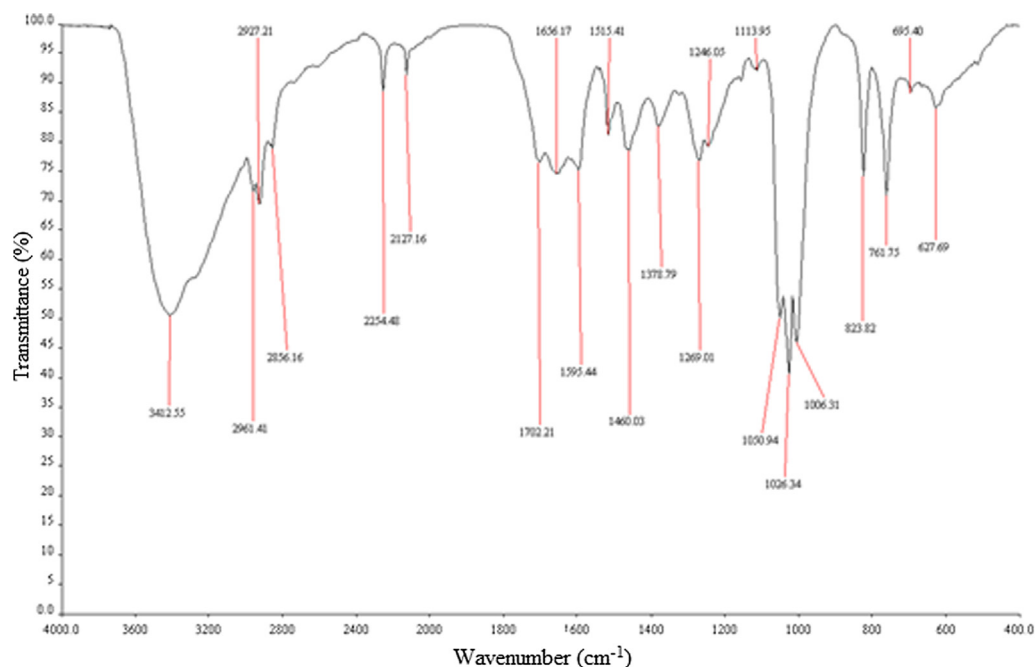
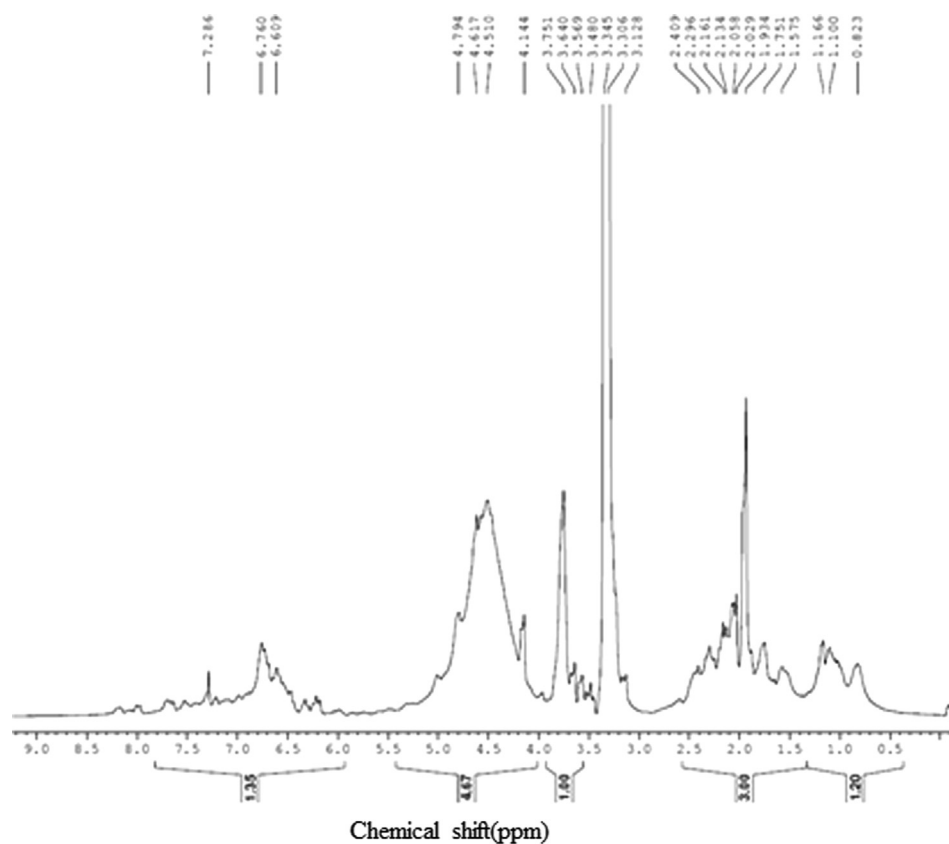


Fig. 7. FT-IR spectrum of bio-oil obtained from sawdust.

Fig. 8. ^1H NMR spectrum of bio-oil.

Similar studies of raw sawdust and bio-char was reported by Kumar et al., Palniandy et al. and Chellapan et al. [8,34,35].

3.6.2. FT-IR analysis

Bio-char was characterized by FT-IR and its spectrum is presented in the Fig. 11. It suggests the presence of functional groups, viz. C-I, C-H, C-O, C = C and O-H. These groups indicate that different classes of

organic compounds such as, halides, phenols, alcohols, alkanes, aromatics, and ketones could be present in it in minor quantities, since the peaks are less intense. Previously Kumar et al., Verma et al., and Morali have analyzed bio-char using FT-IR and reported similar findings [8,10,21].

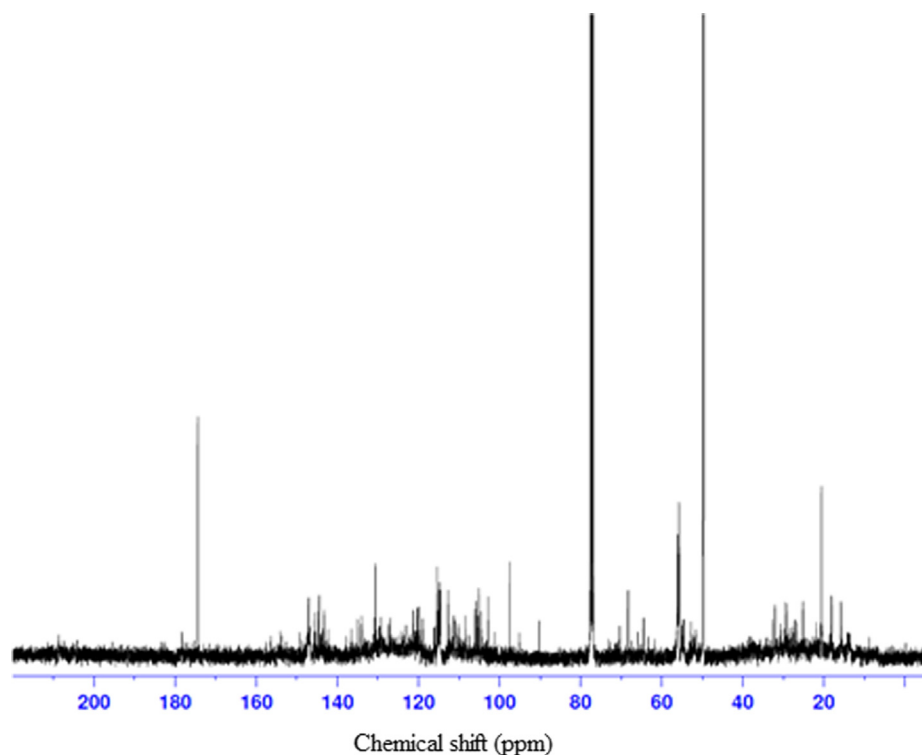


Fig. 9. ^{13}C NMR spectrum of bio-oil.

Table 8
Ultimate analysis of bio-char.

C (wt%)	H (wt%)	O ^a (wt%)	N (wt%)	H/C molar ratio	Empirical equation
74.42	2.97	10.26	1.02	0.5006	$\text{CH}_{0.5006}\text{O}_{0.2178}\text{N}_{0.0118}$

^a By difference.

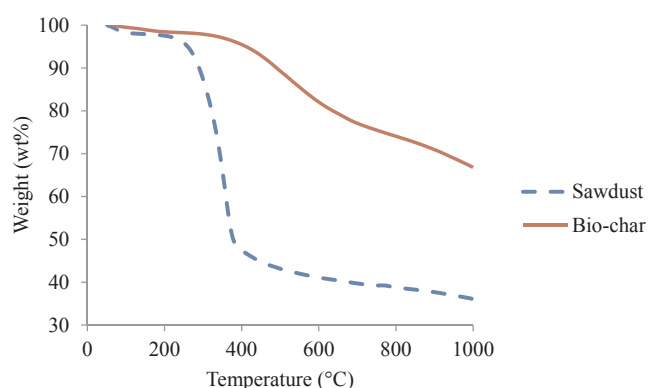


Fig. 10. Comparison of TGA thermograms for sawdust and bio-char.

3.6.3. SEM

Scanning Electron Micrograph (SEM) is an electron beam based microscope and it produces high resolution images of samples by scanning its surface. This scanning gives the output in the form of images which give information about the surface topography and composition of the sample. Magnification range in this microscopy is in the range of several lakhs; it is impactful than normal optical microscope. Both raw sawdust and bio-char were analyzed using SEM. It can be clearly seen that in the SEM image of bio-char, there are holes created on its surface during pyrolysis process of sawdust. This is because of evolution of volatile matters out of biomass during the course of

thermal degradation. Using SEM, several research groups have reported porous structure of sawdust bio-char [8,10,36–38].

3.7. Pyro-gas characterization

GC analysis of pyro-gas was carried out to analyze the composition and concentrations of various individual components. Hydrogen, oxygen, nitrogen, methane, carbon monoxide and carbon dioxide were detected in pyro-gas (Table 9). Pyro-gas generated from sawdust pyrolysis was analyzed by other research groups as well [6,9,10,12–14,16–19]. The reports so far indicate that for each study, pyro-gas composition is different. DeSisto et al. had observed similar concentrations of CO and CH₄ in pyro-gas as reported in this study [13].

4. Conclusion

In this study, valorization of sawdust *via* pyrolysis was carried out in both batch and continuous mode. A pilot scale biorefinery concept is applied to produce bio-oil and bio-char from sawdust. Optimization of the process was carried out for bio-oil production using a batch scale fixed bed reactor. Afterwards, sawdust was pyrolysed in a pilot scale continuous reactor (capacity: 15 kg/hr). Analyses of bio-oil using FT-IR, NMR (^1H and ^{13}C) and GC-MS shows that it is a pool of oxygenated and highly functionalized chemical compounds. These chemical compounds can be separated and purified *via* different downstream processing methods. The purified compounds can be used as fine chemicals for different applications in chemicals and life sciences industries. Images captured using SEM analysis indicated that generated bio-char is porous in nature. Bio-char could be used for purification of water and as a soil amendment.

In the batch pyrolysis bio-oil yield was 34.9 wt% and bio-char yield was 38.6 wt%. Furthermore it was found that the continuous pyrolysis resulted in 41.05 wt% bio-oil and 27.59 wt% bio-char. This process can be further extended for industrial scale bio-oil and bio-char production. Quantification of gaseous components in the pyro-gas was carried out using GC-TCD. It revealed the presence of various gases *viz.* H₂ (1.59 vol

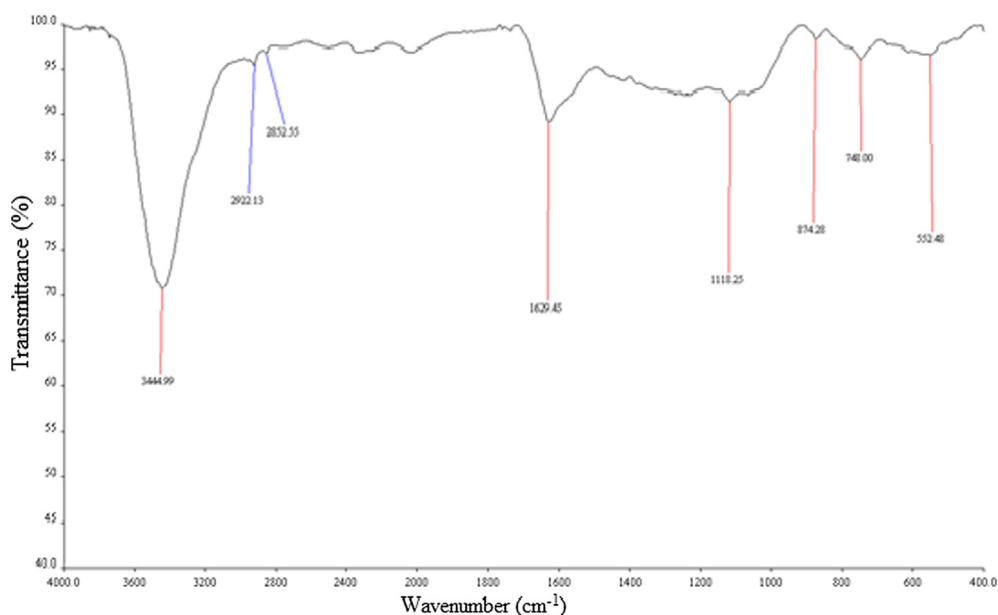


Fig. 11. FT-IR spectrum for bio-char.

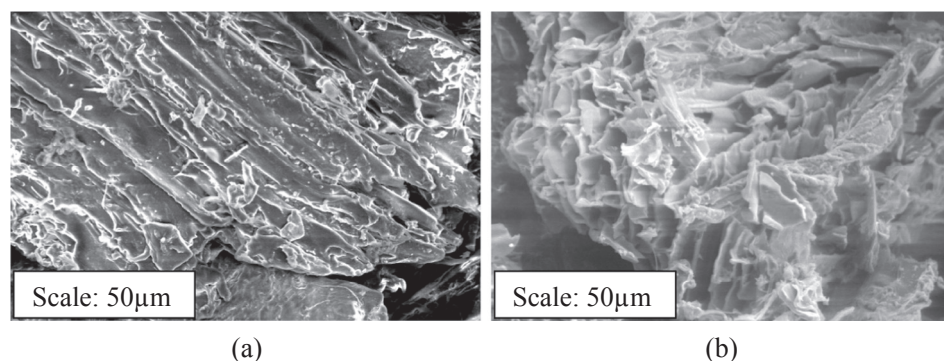


Fig. 12. SEM images (1500x magnification) of sawdust (a) and sawdust bio-char (b).

Table 9
Gas chromatograph analysis of pyro-gas.

Serial no.	Compound name	Retention Time (min)	Gas Composition (Vol. %)
1	Hydrogen (H ₂)	0.965	1.59
2	Oxygen (O ₂)	1.552	2.48
3	Nitrogen (N ₂)	2.282	13.61
4	Methane (CH ₄)	3.643	8.02
5	Carbon monoxide (CO)	6.700	54.23
6	Carbon dioxide (CO ₂)	2.858	24.72

%, N₂ (13.61 vol%), O₂ (2.48 vol%), CO (54.23 vol%), CH₄ (8.02 vol %) and CO₂ (24.72 vol%) in it.

CRediT authorship contribution statement

Bhavin Soni: Investigation, Writing - original draft. **Sanjib Kumar Karmee:** Conceptualization, Supervision, Writing - original draft, Writing - review & editing.

Declaration of Competing Interest

The authors declare that they have no known competing financial

interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.fuel.2020.117570>.

References

- [1] Soni B, Patel U, Mishra G, Karmee SK. Sustainable valorization of sawdust via batch and continuous pyrolysis [abstract]. In: Environmental Pollution & It's Control (EPIC-

- 2019), 2019 Feb 18-19, GEC Banswara, Rajasthan, India. 2019. Abstract no 141 https://www.researchgate.net/publication/332106054_Sustainable_valorization_of_sawdust_via_batch_and_continuous_pyrolysis.
- [2] Energie Ravi. Biomass assessment study of castor de-oiled cake and sawdust based power plant. Ahmedabad, Gujarat: Claris Lifesciences Ltd.; 2019 (accessed on 14.10.19.).
 - [3] Czernik S, Bridgwater AV. Overview of applications of biomass fast pyrolysis oil. *Energy Fuels* 2004;18:590–8. <https://doi.org/10.1021/ef034067u>.
 - [4] Hou SS, Huang WC, Rizal FM, Lin TH. Co-firing of fast pyrolysis bio-oil and heavy fuel oil in a 300-kW_{th} furnace. *Appl. Sci.* 2016;6:326 <https://doi.org/10.3390/app6110326>.
 - [5] Kim E, Gil H, Park S, Park J. Bio-oil production from pyrolysis of waste sawdust with catalyst ZSM-5. *J Mater Cycles Waste Manag* 2017;19:423–31. <https://doi.org/10.1007/s10163-015-0438-z>.
 - [6] Ngo TA, Kim J. Fast pyrolysis of pine wood chip in a free fall reactor: the effect of pyrolysis temperature and sweep gas flow rate. *Energy Sources Part A* 2014;36(11):1158–65. <https://doi.org/10.1080/15567036.2011.631089>.
 - [7] Salehi E, Abedi J, Harding T. Bio-oil from sawdust: pyrolysis of sawdust in a fixed bed system. *Energy Fuels* 2009;23:3767–72. <https://doi.org/10.1021/ef900112b>.
 - [8] Kumar P, Kumar P, Rao PVC, Choudary NV, Sriganesh G. Saw dust pyrolysis: Effect of temperature and catalysts. *Fuel* 2017;199:339–45. <https://doi.org/10.1016/j.fuel.2017.02.099>.
 - [9] Tinwala F, Mohanty Parmar S, Patel A, Pant KK. Intermediate pyrolysis of agro-industrial biomasses in bench-scale pyrolyser: product yields and its characterization. *Bioresour Technol* 2015;188:258–64. <https://doi.org/10.1016/j.biortech.2015.02.006>.
 - [10] Varma AK, Thakur LS, Shankar R, Mondal P. Pyrolysis of wood sawdust: Effects of process parameters on products yield and characterization of products. *Waste Manage* 2019;89:224–30. <https://doi.org/10.1016/j.wasman.2019.04.016>.
 - [11] Yorgan S, Yildiz D, Slow pyrolysis of paulownia wood: Effects of pyrolysis parameters on product yields and bio-oil characterization. *J. Anal. Appl. Pyrolysis* 2015 Doi:10.1016/j.jaap.2015.05.003.
 - [12] Duanguoama K, Suwapaet N, Pattiya A. Fast pyrolysis of contaminated sawdust in a circulating fluidized bed reactor. *J Anal Appl Pyrolysis* 2016;118:63–74. <https://doi.org/10.1016/j.jaap.2015.12.025>.
 - [13] DeSisto WJ, Hill N, Beis SH, Mukkamala S, Joseph J, Baker C, et al. Fast pyrolysis of pine sawdust in a fluidized-bed reactor. *Energy Fuels* 2010;24:2642–51. <https://doi.org/10.1021/ef901120h>.
 - [14] Aguado R, Olazar M, San José MJ, Aguirre G, Bilbao J. Pyrolysis of sawdust in a conical spouted bed reactor. yields and product composition. *Ind Eng Chem Res* 2000;39:1925–33. <https://doi.org/10.1021/ie990309v>.
 - [15] Salehi E, Abedi J, Harding T. Bio-oil from sawdust: effect of operating parameters on the yield and quality of pyrolysis products. *Energy Fuels* 2011;25:4145–54. <https://doi.org/10.1021/ef200688y>.
 - [16] Heo HS, Park HJ, Park YK, Ryu C, Suh YW, et al. Bio-oil production from fast pyrolysis of waste furniture sawdust in a fluidized bed. *Bioresour Technol* 2010;101:S91–6. <https://doi.org/10.1016/j.egypro.2017.07.275>.
 - [17] Olazar M, Akarregi ARF, Makibar J, Lopez G, Amutio M. Design and operation of a conical spouted bed reactor pilot plant (25 kg/hr) for biomass fast pyrolysis. *Fuel Process Technol* 2013;112:48–56. <https://doi.org/10.1016/j.fuproc.2013.02.022>.
 - [18] Park JY, Kim JK, Chang-Ho O, Park JW, Kwon EE. Production of bio-oil from fast pyrolysis of biomass using a pilot-scale circulating fluidized bed reactor and its characterization. *J Environ Manage* 2019;234:138–44. <https://doi.org/10.1016/j.jenvman.2018.12.104>.
 - [19] Jung S, Kang B, Kim JS. Production of bio-oil from rice straw and bamboo sawdust under various reaction conditions in a fast pyrolysis plant equipped with a fluidized bed and a char separation system. *J. Anal. Appl. Pyrolysis* 2008;82:240–7. <https://doi.org/10.1021/ie050486y>.
 - [20] Biomass, Thermo-Chemical Characterization, 3rd Edition, Edited by P.V.R. Iyer, T. R. Rao, P.D. Grower, Published under MNES Sponsored GASIFIER ACTION RESEARCH PROJECT.
 - [21] Morali U, Şensöz S. Pyrolysis of hornbeam shell (*Carpinus betulus* L.) in a fixed bed reactor: Characterization of bio-oil and bio-char. *Fuel* 2015;150:672–8. <https://doi.org/10.1016/j.biortech.2016.09.081>.
 - [22] Gray GG. Determination and Significance of Activation Energy in Permanence Tests. In: John C. Williams (Ed.), *Preservation of Paper and Textiles of Historic and Artistic Value* 1978;164:286–313. <https://dx.doi.org/10.1021/ba-1977-0164.ch020>.
 - [23] Ren S, Lei H, Wang L, Bu Q, Chen S, Wu J. Hydrocarbons and hydrogen rich syngas production by biomass catalytic pyrolysis and bio-oil upgrading over biochar catalysts. *RSC Adv* 2014;4:10731–7 <https://doi.org/10.1039/c4ra00122b>.
 - [24] Zhang M, Shen Q, Wu H. Adsorption characteristics of bio-oil on bio-char in bio-slurry fuels. *Energy Fuels* 2017;31:9619–26 <https://doi.org/10.1021/acs.energyfuels.7b02041>.
 - [25] Lee J, Kim K-H, Kwon EE. Biochar as a catalyst. *Renew Sust Energ Rev* 2017;77:70–9 <https://doi.org/10.1016/j.rser.2017.04.002>.
 - [26] Engineering ToolBox, (2003). Fuels - Higher and Lower Calorific Values. [online] Available at: https://www.engineeringtoolbox.com/fuels-higher-calorific-values-d_169.html [2019]. Accessed on 28/09/2019.
 - [27] Qureshi KM, Faisal Abnisa F, Ashri WM, Daud W. Novel helical screw-fluidized bed reactor for bio-oil production in slow-pyrolysis mode: a preliminary study. *J. Anal. Appl. Pyrolysis* 2019. <https://doi.org/10.1016/j.jaap.2019.04.021>.
 - [28] Sinağ A, Uskan B, Gülbay S. Detailed characterization of the pyrolytic liquids obtained by pyrolysis of sawdust. *J. Anal. Appl. Pyrolysis* 2011;90:48–52. <https://doi.org/10.1016/j.rpor.2010.10.003>.
 - [29] Zhao S, Luo Y, Su Y, Zhang Y, Long Y. Experimental Investigation of the Oxidative Pyrolysis Mechanism of Pinewood on a Fixed-Bed Reactor. *Energy Fuels* 2014;28:5049–56. <https://doi.org/10.1021/ef500612q>.
 - [30] Gao N, Liu B, Li A, Li J. Continuous pyrolysis of pine sawdust at different pyrolysis temperatures and solid residence times. *J. Anal. Appl. Pyrolysis* 2015. <https://doi.org/10.1016/j.jaap.2015.05.011>.
 - [31] Qiang L, Wen-zhi L, Dong Z, Xi-feng Z. Analytical pyrolysis–gas chromatography/mass spectrometry (Py–GC/MS) of sawdust with Al/SBA-15 catalysts. *J. Anal. Appl. Pyrolysis* 2009;84:131–8. <https://doi.org/10.1016/j.jaap.2009.01.002>.
 - [32] Gu X, Ma X, Li L, Liu C, Cheng K, Li Z. Fast pyrolysis of poplar wood sawdust with la-containing sba-15 catalyst by py-gc/ms analysis. *Cellulose Chem Technol* 2014;48(3–4):275–83.
 - [33] Zhang ZZ, Zhu MM, Liu PF, Wan WC, Zhou WX, Chan YL, et al. Effect of biochar on the cracking of tar from the pyrolysis of a pine sawdust in a fixed bed reactor. *Energy Procedia* 2015;75:196–201. <https://doi.org/10.1016/j.egypro.2015.07.299>.
 - [34] Palnandy KL, Yoon LW, Wong WY, Yong ST, Pang MM. Application of biochar derived from different types of biomass and treatment methods as a fuel source for direct carbon fuel cells. *Energies* 2019;12:2477. <https://doi.org/10.3390/en12132477>.
 - [35] Chellappan S, Nair V, Sajith V, Aparna K. Synthesis, optimization and characterization of biochar based catalyst from sawdust for simultaneous esterification and transesterification. *Cjche* 2018.
 - [36] Chowdhury ZZ, Karim MZ, Ashraf MA, Khalid K. Influence of carbonization temperature on physicochemical properties of biochar derived from slow pyrolysis of durian wood (*Durio zibethinus*) sawdust. *BioResources* 2016;11(2):3356–72.
 - [37] Wan A, Wan AB, Ghani K. Sawdust-derived biochar: characterization and CO₂ adsorption/desorption study. *J. Applied Sci.* 2014;14(13):1450–4 <https://doi.org/10.3923/jas.2014.1450.1454>.
 - [38] Liang H, Chen L, Liu G, Zheng H. Surface morphology properties of biochars produced from different feedstocks. 2016. <https://doi.org/10.2991/iccte-16.2016.210>.